

The Economic Impact of CPTPP Membership Evidence from GDP per Capita and Pharmaceutical Patenting

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Introduction

The Comprehensive and Progressive agreement for Trans-Pacific Partnership is a multilateral FTA among 11 countries spanning the Asia-Pacific region, the Americas, and Europe linking the Pacific Rim and Latin American Economies. This agreement was Signed on March 8, 2018, which aims to lower trade barriers, cuts tariffs, and sets shared rules for digital trade, labour, and environmental standard. This agreement is a direct successor of trans pacific partnership and was reconstructed.

The TPP was founded in 2005 by Brunei, Chile, New Zealand and Singapore and was called as the “Pacific-4”. In 2008, the United States became a member of the negotiations and was followed, Australia in 2009, Peru in 2010, Vietnam in 2011, Malaysia in 2012, Canada in 2014, Mexico in 2015 and Japan in 2017, bringing the total number of countries to twelve. The twelve countries have negotiated for over five years and signed the Trans-Pacific Partnership (TPP) on 4 February 2016 in Auckland, New Zealand comprising total of 38% of global GDP at the time.

But, the TPP never came into effect. In January 2017, the incoming Trump administration formally rejected the deal, stating the concerns with job offshoring and inequitable trade. In May 2017, the remaining eleven members led by Japan, agreed to revive a modified version of the agreement. The updated agreement was signed in Santiago, Chile on 8 March 2018, is now known as the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP). This is one key element of this renegotiation that is central to this study: The 11 remaining members omitted 22 of more than 600 provisions in the original TPP text. The vast majority of these conditions were on intellectual property rights, specifically pharmaceutical patent protection that the United States had insisted on and was not supported by other member states. The CPTPP formally came into force on 30 December 2018 for the first six countries to sign it (Australia, Canada, Japan, Mexico, New Zealand, Singapore), followed by Vietnam in 2019, Peru in 2021, Malaysia and Chile in 2022 and Brunie in 2023. CPTPP has been ratified by all original CPTPP signatories.

Strategic Importance

The CPTPP represents the 13% of the global GDP accounting for more than USD 10 trillion. It spans a market of approximately 500 million consumers. From a strategic standpoint, the agreement plays a major role in the geopolitics.

It preserved the Asia Pacific high standard rules-based trading system at a time when the United States withdrew its leadership from multilateral trade exhibiting the stability of trade. It acts as a counterweight and alternative pole to Asia-centred regional blocs like RCEP, thereby allowing its members to access a diversified market. It is also one of the most diverse acceding trade blocs in the world, with a wide variety of new members joining, including the United Kingdom (acceded 2024), and has a pending membership application from China, Taiwan, South Korea, Ecuador, Costa Rica and Ukraine.

CPTPP membership provides preferential access to large markets that traditionally have been hard to negotiate in bilateral agreements, such as Japan and Canada, for smaller or trade-dependent economies in the group (Vietnam, Malaysia, Peru, etc) thus improving trade openness.

Research Question

This project aims to derive the measurable economic effect of CPTPP in its member states with dual perspectives.

1. Did CPTPP membership cause a change in GDP per capita among the member countries?
2. Did CPTPP membership cause a change in pharmaceutical patent grants among member countries, given the changes introduced in the agreement's intellectual property provisions?

These two research questions have been chosen as GDP serves as a broad parameter to test the translation of trade liberalisation into aggregated economic gains and pharmaceutical patents serve as a more targeted test.

Intellectual Property Chapter- Change from TPP to CPTPP

The renegotiation of the TPP into CPTPP brought in multiple changes in the earlier provisions out of which 22 provisions were suspended. The substantial number of suspensions were related to Pharmaceutical Intellectual property rules. TPP had introduced very lenient patent rules to promote the patents which was highly supported by US. CPTPP suspended multiple articles related to Patent-term adjustment and patent-term restoration for regulatory, protection of undisclosed test data for both small-molecule drugs and biologics and Expansion of patentable subject matter to cover new uses. Thus, making the patent rules stricter. This drives the question whether new rules lead to any measurable change in the patent grants in the CPTPP countries? This question is tried to solve in this analysis.

Data and Variables

The country-year panel dataset from 2005 to 2024, with 39 variables and 26 countries is built. 11 CPTPP members (Australia, Brunei, Canada, Chile, Japan, Malaysia, Mexico, New Zealand, Peru, Singapore, and Vietnam) and 15 non-member country (Italy, India, Spain, Poland, Brazil, Argentina, Thailand, Colombia, Indonesia, Philippines, Germany, South Korea, France, United Kingdom) is used for the analysis. The control variables were chosen based on geographical and economic overlaps. Most of the control variables are the neighbouring countries of CPTPP

member states. The advanced countries which have higher patents are also used as control variables for a better comparison in pharmaceutical analysis, using the public API of WDI, macroeconomic and health control variables were compiled namely GDP per capita, population, share of merchandise trade to GDP, foreign direct investment inflows, health expenditure per capita, R&D expenditure as a share of GDP, researchers per million people, scientific journal articles, and high-technology exports. Second, the main outcome measures pharmaceutical patent grants and biotechnology patent grants were extracted from the WIPO IP Statistics Data Centre. Third, the number of European patent applications filed in the same timeframe and by the applicant origin was retrieved from the EPO Technology Dashboard 2025. In addition to sources from the World Bank's IP indicators.

Methodology

Difference in Difference causal method is used to measure the impact of CPTPP. DiD is the natural design that isolates the policy's causal effect by comparing the change in outcomes for the treated countries (members of the agreement) before and after joining against the change over the same period for untreated countries (countries used for the comparison i.e. Control variables). It therefore removes the effects of fixed differences and common trends affecting all countries. The two period Did regression takes the following form.

$$Y_{it} = \beta_0 + \beta_1 \cdot \text{Treat}_i + \beta_2 \cdot \text{Post}_t + \beta_3 \cdot (\text{Treat}_i \times \text{Post}_t) + \varepsilon_{it}$$

In this particular analysis the CPTPP was not adopted on a single date instead various members ratified the agreement in various years such as 2018, 2019, 2021, 2022, and 2023. A standard two-period DiD, or even a two-way fixed-effects (TWFE) regression using a single shared cutoff date, is known in the econometrics literature to produce biased estimates under staggered adoption. Therefore, the Callaway and Sant'anna Estimator is used.

It assigns a group (cohort) for each treated country that is determined by the actual year of ratification, instead of adopting a common treatment year imposed on all. It calculates the impact of each cohort by including only “not-yet-treated” countries as the valid comparison group at each time horizon. Thus, it eliminates the potential source of TWFE bias that would arise in the case of staggered adoption. It enables one to observe pre-treatment estimates for each cohort separately with a transparent parallel trends' diagnostic, instead of a single pooled assumption.

$$ATT(g,t) = E[Y_t - Y_{\{g-1\}} | G_g = 1] - E[Y_t - Y_{\{g-1\}} | C = 1], \text{ for } t \geq g$$

where $G_g = 1$ represents the members of the cohort treated in period g , $C = 1$ represents the comparison (not-yet-treated) group, and $Y_{\{g-1\}}$ is the outcome before the cohort's treatment

$ATT(g,t)$ is the average change in the outcome for cohort g between its pre-treatment baseline and period t , minus the corresponding change for units not yet treated. The standard DiD logic is used but applied separately, to every cohort and every time period.

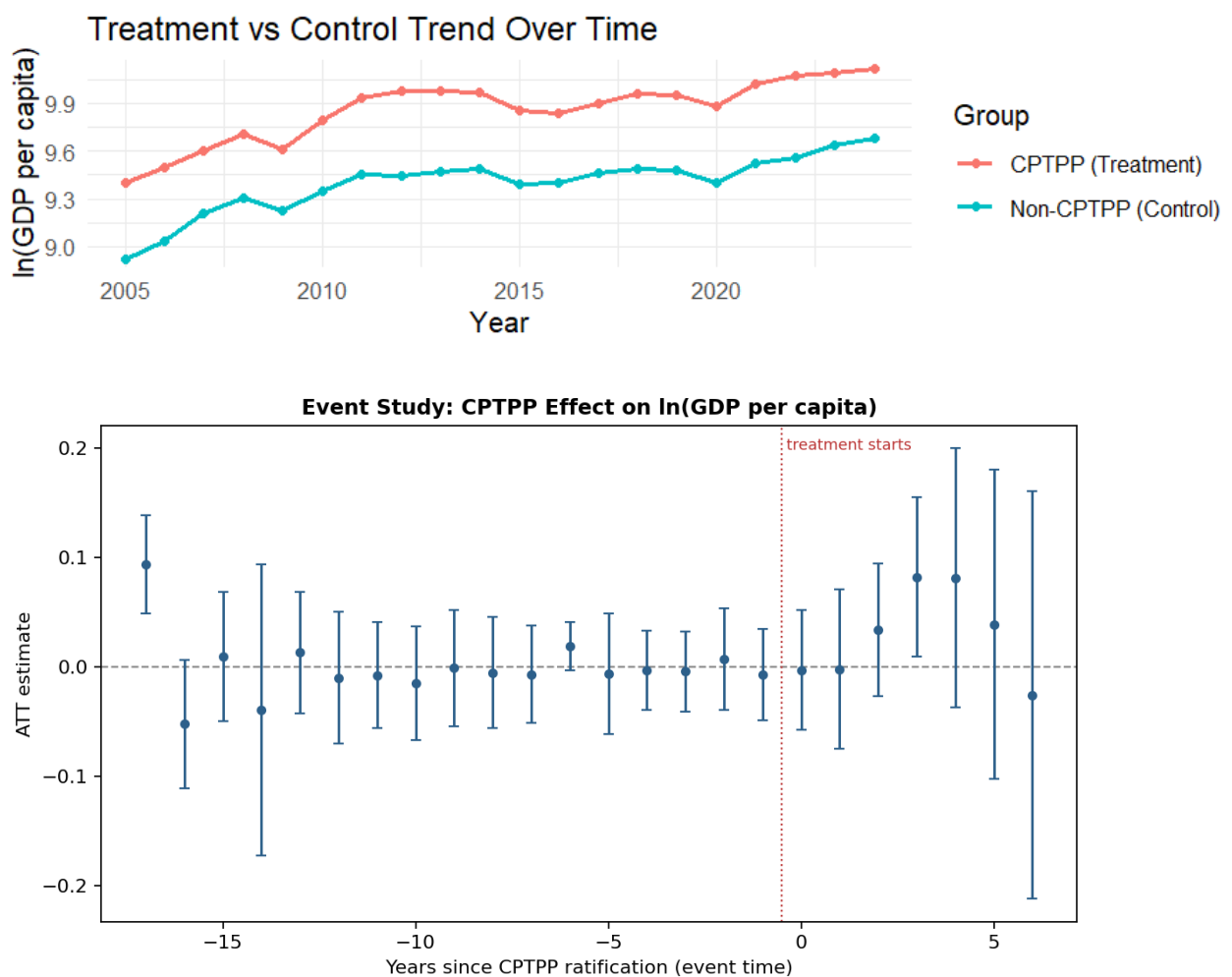
The group-time effects are then aggregated in a number of ways such as an overall average treatment effect on the treated (ATT), a dynamic (event-study) aggregation by time since

treatment, $e = t - g$ (years since treatment), and an aggregation by group (group effect) reporting one effect per set of ratifications.

Results

GDP Analysis

The parallel trend test prove that the GDP of Control and Treatment Countries before the CPTPP followed the same pattern thus the Parallel Trend exists. The Treatment variables naturally have higher aggregate GDP than the Control Variables (figure 1). The pretreatment estimates for the 2018 were uniformly statistically insignificant across all pre-treatment years. The 2019 cohort (Vietnam), show statistically significant positive pre-treatment ATT(g,t) values (years 2008–2010, 2012–2015), indicating that this cohort was already on a faster GDP growth trajectory before joining CPTPP thus does not follow parallel trend. Post Treatment ATT(g,t) for GDP is insignificant for every cohort. ATT in simple analysis is 0.0273 and dynamic ATT is 0.0297(figure 2).



Interpretation - No statistically detectable effect of CPTPP membership on GDP per capita. Standard errors exceed point estimates throughout, consistent with limited statistical power (only 5 treated cohorts, several with very short post-treatment windows) rather than conclusive evidence of a true zero effect.

Pharmaceutical Patent Analysis Results

Several individual ATT(g,t) cohorts show statistical significance in both pre and post treatment.

2019 cohort Vietnam shows strongly significant post-treatment effects at 2020 (0.90), 2022 (1.29), 2023 (0.91), and 2024 (1.96) all of them being large and positive. This shows that post treatment periods showed steep increase in the Pharma Patents in Vietnam. But the overall aggregate CPTPP affect on all the members show insignificant results. This is a clear sign of active building of pharmaceutical manufacturing and IP infrastructure over this period, partly independent of CPTPP. The positive effect is real but the causation from the CPTPP is unclear.

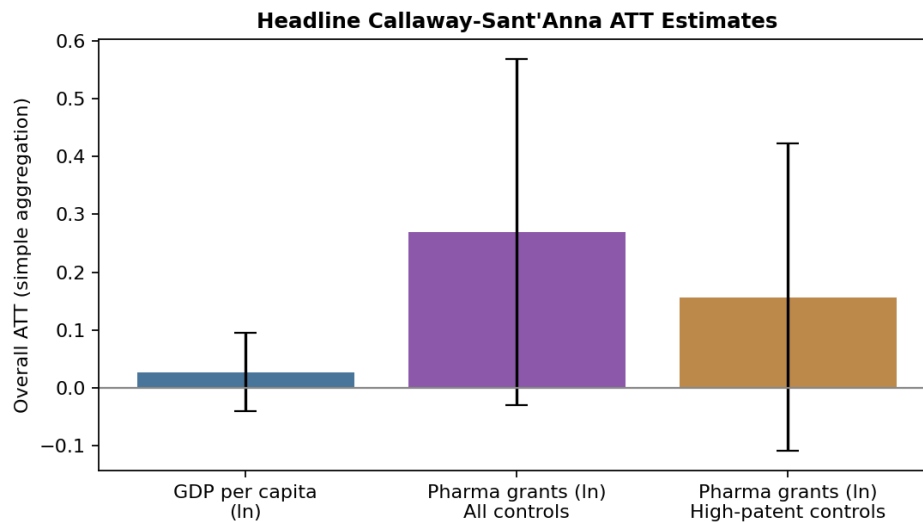
2021 cohort (Peru) shows significant but sign inconsistent pre-treatment cells (+0.58 at 2011, -0.91 at 2012) alongside significant post-treatment cells such as +0.77 (2018) and -0.57 (2019). This result is a strong signal that Peru's pharma patent counts are simply low-volume and noisy. This signals that additional grants in a small country creates a huge increase in terms of percentage in log terms.

2022 cohort (Malaysia) and 2023 cohort (Brunei, Chile) signs numerous significant cells both before and after treatment, with unstable sign in adjacent years (e.g., the 2023 cohort moves from +0.44 to -0.84 to +0.62 across three consecutive pre-treatment years). The drastic change in signs in a small period of time showcases the volatility created by data insufficiency. Thus, the results should not be read as a result of CPTPP.

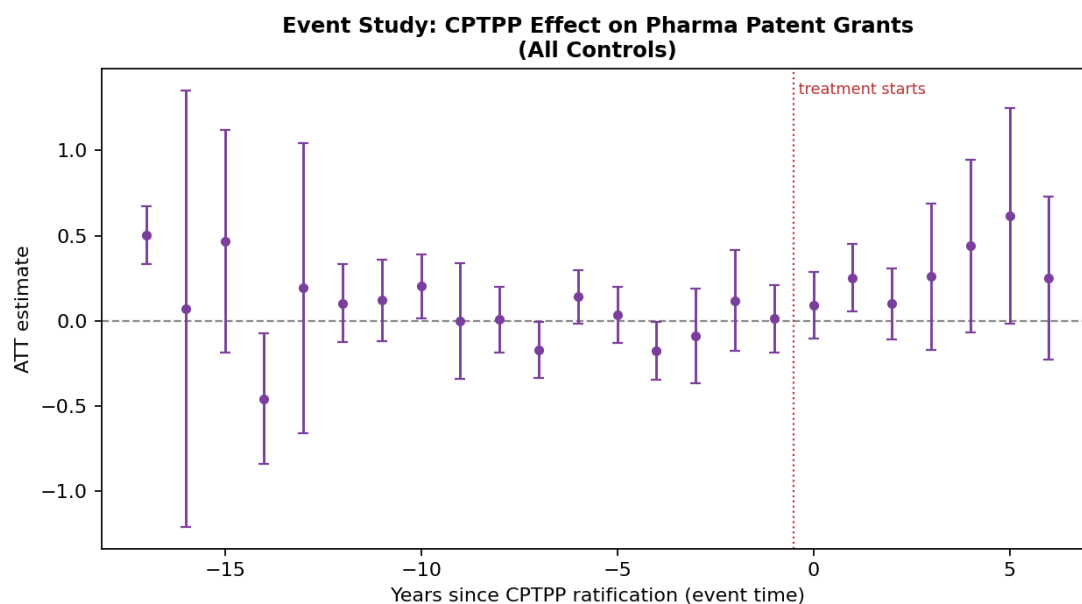
2018 cohort (6 countries) is a cleanest cohort of the analysis shows zero significant ATT(g,t) cells anywhere, pre- or post-treatment. This tells that for a best identified group there is no evidence of any changes in the Patent Pharma Sector caused by CPTPP.

The aggregate results show ATT of 0.269 with SE of 0.153. The results are significant at 10% with P value being 0.079. Thus, the significance is at borderline. This suggests that CPTPP membership is associated with an estimated 30.9% increase in pharmaceutical patent grants (95% CI: -3.0% to +76.5%). Under the stricter, better-matched control group, the estimated effect falls to a 16.9% increase (95% CI: -10.4% to +52.5%), and is not statistically significant.

To check if the caveat in the analysis is created by insufficient data, the Callaway and Sant'anna DiD was extended only to the countries which have large datasets. Even high Patent controls showed insignificant results thus proving no major effect of CPTPP in Pharma Patent Sector.



This graph shows Headline ATT estimates across various outcomes and specifications. The vertical bars show 95% Confidence Interval.



The overall ATT using all available controls (0.269) sits just above conventional significance (95% CI lower bound -0.030) it is thus significant at 10%. The dynamic event-study shows a rising post-treatment pattern from event time +1 through +5 (0.25 to 0.61), though only event time +1 is individually significant.

Conclusion

The simple and the dynamic aggregate ATTs are small and statistically insignificant, and their standard errors are large, and the result on GDP per capita is a clean null: neither are large in absolute terms nor large relative to their standard errors.

The project's more novel findings on pharmaceutical patents appeared as a potentially positive sign of impact an aggregate effect of 30% with its results being significant at the 10% level. Most importantly, the single cohort identified best (2018, six country samples, clean pre-trends)

reports no significant effect anywhere, and the positive overall effect is largely attributable to small, low patent-count countries (Vietnam, Peru, Malaysia, Brunei, Chile) where the sign of few countries shows the flipping from one year to the next at neighbouring time points, suggesting statistical noise rather than a true treatment effect.

The difference between the two results is informative. The impact of a trade agreement on tariffs and market access is an offset by the macroeconomic noise, driven by dozens of factors, such as domestic fiscal policy, commodity cycles, exchange rates and global demand shocks (including shocks from the COVID-19 crisis which falls clearly into the range of post-treatment years of several trade agreements, and short relative to the period required for trade-driven growth gains to accumulate). Pharmaceutical patent grants, on the other hand, is a more narrow and mechanically related result. The impact of IPR on Patents will be felt quicker. The impact of CPTPP will be first on patents, and last in GDP. The impact of CPTPP doesn't even hold for patents with stricter controls. Thus, this provides a stronger basis for scepticism about the agreement's short-run economic impact along with the GDP

The patent cohorts in this group also directly overlap with the COVID-19 pandemic, which saw a spike in patent filing activity in 2020-2021 for vaccines and therapeutics, which was unprecedented and correlated with the pandemic across the world which can lead to an artificial surge in post-treatment patent count. This is a problem only if the pandemic had a disproportionate impact on patenting activity in CPTPP member countries relative to non-CPTPP countries (e.g., if pandemic related patenting activity was higher in Vietnam, which has a more mature pharmaceutical manufacturing industry). TRIPS flexibility, the initiatives on generic drugs and vaccines promoted by the WHO, or unrelated bilateral or multilateral IP agreements that individual member countries may have negotiated during the CPTPP period, may have influenced patent filing incentives independently of the IP effect from the CPTPP itself.

The project's overall conclusion is that CPTPP membership shows no credible, robust evidence of causally improving GDP per capita or pharmaceutical patent activity among its members over the 2005–2024 study period.

